Lab Report – EE-170L Computer Programming (Lab 10)

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DATE	5/5/2026

Objectives

- *Understand **function call by value**.*
- *Understand **function call by reference**.*
- *Learn about **default parameter values** in functions.*
- *Apply these concepts in practical C++ programs.*

Theory

***Call by Value**: A copy of the variable is passed to the function. Changes inside the function do not affect the original variable.*

***Call by Reference**: The actual memory address is passed. Changes inside the function directly affect the original variable.*

***Default Parameters**: Functions can have default values for parameters, which are used if no argument is provided during the call.*

Lab Tasks

Task 1: Function with Parameters

```
#include <iostream>                // include input/output library
using namespace std;               // use standard namespace

void multiply(int num1, int num2) {  // function with two integer parameters
    int product = num1 * num2;         // calculate product of num1 and num2
    cout << "Product = " << product;   // display result inside function
}
```

```
int main() {                        // main function starts
    multiply(6, 7);                 // call multiply function with two numbers
    return 0;                       // end program
}
```

Task 2: Function with Return Value

```
#include <iostream>                 // include input/output library
using namespace std;                // use standard namespace

int getSquare(int number) {             // function with one integer parameter
    int square = number * number;       // calculate square of the number
    return square;                      // return the result
}

int main() {                            // main function starts
    int result = getSquare(9);          // call function and store returned value
    cout << "Square = " << result;      // display the square
    return 0;                           // end program
}
```

Results

- *Task 1 displayed:* ***Product = 42*** *(for inputs 6 and 7).*
- *Task 2 displayed:* ***Square = 81*** *(for input 9).*

Conclusion

In this lab, I learned:

- *How to pass parameters to functions.*
- *How to return values from functions.*
- *The difference between call by value and call by reference.*
- *How default parameters simplify function calls.*

*Functions make programs **modular, reusable, and efficient**.*